

- Description

This module manages a smart boiler temperature control system, ensuring that:

1. The system continuously monitors the current boiler temperature (in °C).
2. If the temperature falls below a predefined minimum threshold, the heating element is activated to increase the temperature.
3. If the temperature exceeds a predefined maximum threshold, the heating element is deactivated to prevent overheating.
4. If the temperature is within the optimal range, no action is taken.

This system helps maintain an optimal water temperature in the boiler, ensuring energy efficiency and preventing damage due to overheating or freezing.

- Specification

module SmartBoilerTemperatureControl

var currentTemperature, minTemperature, maxTemperature: real

var heatingActive: boolean

process updateTemperature(newTemperature: real) returns (currentTemperature: real, heatingActive: boolean)

functional scenario form:

currentTemperature < minTemperature && heatingActive = true, output "Low temperature detected: " + currentTemperature + "°C. Heating activated.", Heating activated.

||

currentTemperature > maxTemperature && heatingActive = false, output "High temperature detected: " + currentTemperature + "°C. Heating deactivated.", Heating deactivated.

||

currentTemperature <= maxTemperature && heatingActive = false, output "Temperature optimal: " + currentTemperature + "°C. Heating remains off.", Heating remains off.

end process

end module

- Java Code

```
package code;
public class c93_SmartBoiler {
    private double minTemperature;
    private double maxTemperature;
    private double currentTemperature;
    private boolean heatingActive;
    public c93_SmartBoiler(double minTemperature, double maxTemperature
, double initialTemperature) {
        this.minTemperature = minTemperature;
        this.maxTemperature = maxTemperature;
    }
}
```

```

        this.currentTemperature = initialTemperature;
        this.heatingActive = false;
    }
    public void updateTemperature(double newTemperature) {
        currentTemperature = newTemperature;
        adjustHeating();
        System.out.printf("Current Boiler Temperature: %.2f°C, Heating:
%s%n",
                        currentTemperature,
                        heatingActive ? "ON" : "OFF");
        if (heatingActive) {
            System.out.println("ALERT: Low temperature detected. Activating heating element.");
        } else {
            System.out.println("Temperature is optimal or too high. Heating element deactivated.");
        }
    }
    private void adjustHeating() {
        if (currentTemperature < minTemperature) {
            heatingActive = true;
        } else if (currentTemperature > maxTemperature) {
            heatingActive = false;
        }
    }
    public double getCurrentTemperature() {
        return currentTemperature;
    }
    public static void main(String[] args) {
        c93_SmartBoiler boiler = new c93_SmartBoiler(40.0, 80.0, 60.0);
        // testcase-VT:
        //     boiler.updateTemperature(35.0);
        //     boiler.updateTemperature(85.0);
        //     boiler.updateTemperature(60.0);

        // testcase-FT:
        //     boiler.updateTemperature(0.0);
        //     boiler.updateTemperature(-1.7976931348623157E308);
        //     boiler.updateTemperature(4.590432711929855E293);

        // testcase-Z3:
        //     boiler.updateTemperature(35.0);
        //     boiler.updateTemperature(85.0);
        //     boiler.updateTemperature(60.0);
    }
}

```

```

// testcase-UVT:
boiler.updateTemperature(35.0);
boiler.updateTemperature(85.0);
boiler.updateTemperature(60.0);

}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c93_SmartBoiler.java	100.0 %	8	0	8
▼ c93_SmartBoiler	100.0 %	8	0	8
main(String[])		0	0	0
c93_SmartBoiler(double, double)		0	0	0
adjustHeating()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,588	21,677
▼ src	0.4 %	89	21,588	21,677
▼ code	0.4 %	89	21,588	21,677
▼ c93_SmartBoiler.java	96.7 %	89	3	92
▼ c93_SmartBoiler	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c93_SmartBoiler(double, double)	100.0 %	15	0	15
adjustHeating()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c93_SmartBoiler.java	87.5 %	7	1	8
▼ c93_SmartBoiler	87.5 %	7	1	8
main(String[])		0	0	0
c93_SmartBoiler(double, double)		0	0	0
adjustHeating()	75.0 %	3	1	4
updateTemperature(double)	100.0 %	4	0	4

5. Fuzz testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,588	21,677
▼ src	0.4 %	89	21,588	21,677
▼ code	0.4 %	89	21,588	21,677
▼ c93_SmartBoiler.java	96.7 %	89	3	92
▼ c93_SmartBoiler	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c93_SmartBoiler(double, double)	100.0 %	15	0	15
adjustHeating()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c93_SmartBoiler.java	100.0 %	8	0	8
▼ c93_SmartBoiler	100.0 %	8	0	8
adjustHeating()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4
main(String[])		0	0	0
c93_SmartBoiler(double, double, dou		0	0	0

7. Z3 instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,588	21,677
▼ src	0.4 %	89	21,588	21,677
▼ code	0.4 %	89	21,588	21,677
▼ c93_SmartBoiler.java	96.7 %	89	3	92
▼ c93_SmartBoiler	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c93_SmartBoiler(double, double)	100.0 %	15	0	15
adjustHeating()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c93_SmartBoiler.java	100.0 %	8	0	8
▼ c93_SmartBoiler	100.0 %	8	0	8
adjustHeating()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4
main(String[])		0	0	0
c93_SmartBoiler(double, double, dou		0	0	0

9. Unimproved V-method instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,588	21,677
▼ src	0.4 %	89	21,588	21,677
▼ code	0.4 %	89	21,588	21,677
▼ c93_SmartBoiler.java	96.7 %	89	3	92
▼ c93_SmartBoiler	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c93_SmartBoiler(double, double)	100.0 %	15	0	15
adjustHeating()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37